Hydrotreating unit (U-200) operating manual

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1. Purpose and scope

This manual describes how the hydrotreating unit U-200 works and how to operate it: normal operation, start-up, shutdown and the most common upsets. It covers the areas Reaction, Recycle, Separation, Compression, Hydrotreating and Quality. The control loops are described in detail in the control narratives (DOC-CN-001). Equipment data are in DOC-DS-001.

Values in this manual (setpoints, limits) are those of the plant's training simulator, which the PlantCraft demonstration reads over OPC UA.

2. Process description

The hydrotreater removes sulfur and nitrogen from a distillate cut by reacting it with hydrogen over a catalyst. The design capacity is 4,500 m³/d.

The charge, controlled by FIC-1201, is mixed with recycle hydrogen and heated before it enters the hydrotreating reactor R-1203. The reaction is exothermic: the catalyst bed temperature (TIC-1203) rises along the bed. Cold quench (FIC-1206) is injected between the beds to limit the temperature rise.

The reactor effluent is cooled and goes to the high-pressure separator V-1204, where hydrogen-rich gas separates from the liquid. The level of V-1204 is controlled by LIC-1204. The gas goes to the compressor suction drum V-1208 (level LIC-1208) and to the recycle gas compressor K-1103, which returns it to the reactor inlet. The hydrogen purity of the recycle gas is measured by AIC-1205.

The liquid goes to the product separator V-1004, where pressure (PIC-1002), level (LIC-1004) and the oil/water interface (LIC-1010) are controlled, and then to the surge drum V-1105 (LIC-1105). The recycle pump P-1003 returns part of the product to the reactor R-1005 under flow control FIC-1003. The reactor R-1005 temperature is controlled by TIC-1005.

Product quality is followed by three analyzers: product composition (AIC-1006), stream composition (AIC-1108) and sulfur content in product (AIC-1215).

3. Main equipment

Tag	Description	Key data
R-1005	Reactor	120 m³, design 95 bar g / 420 °C
R-1203	Hydrotreating reactor	95 m³, design 110 bar g / 430 °C, two catalyst beds with quench between them

Tag	Description	Key data
V-1204	High-pressure separator	70 m³, design 105 bar g / 90 °C
V-1004	Product separator	60 m³, design 12 bar g / 180 °C
V-1105	Surge drum	45 m³, design 8 bar g / 150 °C
V-1208	Compressor suction drum	18 m³, design 40 bar g / 80 °C
K-1103	Recycle gas compressor	centrifugal, 2,800 kW
P-1003	Recycle pump	centrifugal, 140 m³/h rated, 75 kW

4. Control philosophy

Loop	Controls	Normal setpoint	Criticality
FIC-1201	Hydrotreater charge flow	64 m³/h	medium
TIC-1203	Catalyst bed temperature	66 °C	high
FIC-1206	Quench flow	47 m³/h	medium
PIC-1202	Reactor inlet pressure	56 bar	high
LIC-1204	High-pressure separator level	51 %	medium
AIC-1205	Hydrogen purity	58 mol%	medium
PIC-1103	Compressor suction pressure	46 bar	high
PIC-1207	Compressor discharge pressure	83 bar	medium
LIC-1208	Suction drum level	50 %	medium
PIC-1002	Product separator pressure	52 bar	high
LIC-1004	Product separator level	85 %	medium
LIC-1010	Product separator interface level	51 %	medium
LIC-1105	Surge drum level	50 %	medium
FIC-1003	Recycle flow	58 m³/h	high
TIC-1005	Reactor temperature	62 °C	high
AIC-1006, AIC-1108, AIC-1215	Product quality	50 mol%, 55 mol%, 45 ppm	medium

The catalyst bed temperature TIC-1203 is the most important variable of the unit. It is controlled by the reactor inlet heater, with the quench FIC-1206 as the second handle. A charge increase raises the bed temperature a few minutes later, so charge changes are made in steps of no more than 5 m³/h, waiting for the bed temperature to settle between steps.

5. Operating limits

Variable	Low alarm	Normal	High alarm	Trip or action
Catalyst bed temperature (TIC-1203)	60 °C	66 °C	72 °C	78 °C: depressurize the reactor loop (emergency procedure 8.1)
Reactor temperature (TIC-1005)	56 °C	62 °C	68 °C	74 °C: cut the charge
Reactor inlet pressure (PIC-1202)	50 bar	56 bar	62 bar	relief valve set at the design pressure
Product separator pressure (PIC-1002)	47 bar	52 bar	57 bar	
Product separator level (LIC-1004)	20 %	85 %	95 %	98 %: high-level trip closes the inlet
High-pressure separator level (LIC-1204)	25 %	51 %	80 %	10 %: low-level trip protects against gas blow-by
Compressor suction drum level (LIC-1208)	15 %	50 %	70 %	85 %: compressor trip (liquid carry-over)
Hydrogen purity (AIC-1205)	52 mol%	58 mol%		below 50 mol%: increase purge
Recycle flow (FIC-1003)	40 m³/h	58 m³/h	75 m³/h	

The product separator level normally runs high (85 %) because the outlet valve LV-1004 is at its upper limit during high throughput (see 9.3). Operators watch the high alarm at 95 % closely.

6. Start-up (summary)

1. Confirm the unit is purged with nitrogen and the oxygen content is below 0.5 %.

2. Pressurize the reactor loop with hydrogen to 30 bar. Start the recycle gas compressor K-1103 with PIC-1103 in automatic.
3. Establish the liquid circulation: start P-1003, put FIC-1003 in automatic at 40 m^3/h, then raise to 58 m^3/h.
4. Heat the reactor inlet at no more than 25 °C per hour until the bed temperature reaches 60 °C.
5. Introduce the charge on FIC-1201 at 50 % of the normal rate. Raise it in steps of 5 m^3/h, waiting for TIC-1203 to settle at each step.
6. Put TIC-1005 and TIC-1203 in automatic only after the temperatures are stable within 2 °C of setpoint.
7. Line up the products to storage when AIC-1215 is below 50 ppm.

7. Normal shutdown (summary)

1. Reduce the charge on FIC-1201 in steps of 5 m^3/h to 50 %.
2. Reduce the reactor inlet temperature at no more than 25 °C per hour.
3. Stop the charge. Keep hydrogen circulation for at least 2 hours to strip hydrocarbons from the catalyst.
4. Stop P-1003. Depressurize the reactor loop to the flare at no more than 7 bar per minute.

8. Emergency procedures

8.1 High catalyst bed temperature (temperature runaway)

If TIC-1203 rises above 72 °C and keeps rising: open the quench FIC-1206 fully, cut the charge on FIC-1201, and reduce the reactor inlet temperature. If the bed reaches 78 °C, depressurize the reactor loop to the flare. Never close the quench while the bed temperature is rising.

8.2 Recycle gas compressor trip

A trip of K-1103 stops the hydrogen flow through the reactor. Cut the charge immediately, as the bed temperature rises quickly without hydrogen. Check the suction drum level LIC-1208: liquid carry-over is the most common cause of a trip.

9. Common upsets and what to do

9.1 Recycle flow FIC-1003 jumps and cycles

The recycle flow FIC-1003 cycles with a saw-tooth output and steps in the flow. This is the known stiction of the valve FV-1003 (graphite packing, see DOC-TG-001 section 4.1 and the maintenance history DOC-MH-001). The cycle upsets the reactor R-1005 temperature. Do not raise the controller gain. If the cycle becomes larger than ±5 m^3/h, put FIC-1003 in manual at the last stable output and call instrument maintenance.

9.2 Product separator pressure PIC-1002 oscillates

PIC-1002 has oscillated with a period of about 3 minutes since its controller gain was raised in the 2025 turnaround (INC-2026-007). The oscillation passes to the separator level and to the

downstream units. The control engineer has proposed to reduce the gain by 30 %. Until then, operators may put PIC-1002 in manual for a few minutes to stop a growing cycle.

9.3 Product separator level LIC-1004 at the top

At high throughput the outlet valve LV-1004 is fully open and the level rises toward the high alarm. Reduce the charge by 5 m³/h and inform the shift supervisor. The valve was sized for the original capacity. A larger valve is planned for the next turnaround.

9.4 Reactor temperature TIC-1005 in manual

Operators have kept TIC-1005 in manual during feed changes since the overshoot event of 2025 (INC-2025-031). The loop was retuned after that event and is expected to work in automatic. Agree with the shift supervisor before returning it to automatic, and watch the first feed change.

9.5 Noisy analyzers

AIC-1205 (hydrogen purity) and the quality analyzers AIC-1006, AIC-1108 and AIC-1215 have noisy readings caused by their sample systems. A single high reading is not a reason to change the operation. Wait for three consecutive analyzer cycles or ask for a lab sample.